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Fruit detection, yield prediction and canopy geometric characterization using LiDAR with forced air flow

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Abstract

Yield monitoring and geometric characterization of crops provide information about orchard variability and vigor, enabling the farmer to make faster and better decisions in tasks such as irrigation, fertilization, pruning, among others. When using LiDAR technology for fruit detection, fruit occlusions are likely to occur leading to an underestimation of the yield. This work is focused on reducing the fruit occlusions for LiDAR-based approaches, tackling the problem from two different approaches: applying forced air flow by means of an air-assisted sprayer, and using multi-view sensing. These approaches are evaluated in fruit detection, yield prediction and geometric crop characterization. Experimental tests were carried out in a commercial Fuji apple (*Malus domestica* Borkh. cv. Fuji) orchard. The system was able to detect and localize more than 80% of the visible fruits, predict the yield with a root mean square error lower than 6% and characterize canopy height, width, cross-section area and leaf area. The forced air flow and multi-view approaches helped to reduce the number of fruit occlusions, locating 6.7 % and 6.5 % more fruits, respectively. Therefore, the proposed system can potentially monitor the yield and characterize the geometry in apple trees. Additionally, combining trials with and without forced air flow and multi-view sensing presented significant advantages for fruit detection as they helped to reduce the number of fruit occlusions.

Keywords: Apple detection; Fruit counting; Yield prediction; 3D plant modeling; Geometric characterization

Nomenclature

af	Trial scanning with forced air flow
D	Number of fruit detections
DR	Detection rate
E	Trial scanning from the east side
FoV	Field-of-view
FP	False positive detection
FPR	False positive rate
FPr	False positive removal step
FS	Fruit separation step

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31	GT_{field}	Number of apples manually counted in the orchard
32	GT_{labels}	Number of apples labelled in the 3D point cloud
33	$H1$	Trial scanning with LiDAR sensor at 1.8m height
34	$H2$	Trial scanning with LiDAR sensor at 2.5m height
35	IMU	Inertial measurement unit
36	K	Number of apples in a cluster of 3D points
37	LD	Number of labels detected
38	MD	Multi-detection
39	MDR	Multi-detection rate
40	n	Trial scanning without forced air flow application
41	P	Precision
42	Pp	Pre-processing
43	Pt	Number of points that contain a cluster
44	R	Recall
45	$\bar{R}$	Mean reflectance of the points of a cluster
46	rh	Reflectance histogram
47	$RMSE$	Root mean square error
48	RTK-GNSS	Real-time kinematics global navigation satellite system
49	SVM	Support-vector-machine
50	T	Total number of fruits in the dataset
51	TP	True positive
52	V	Volume of a cluster
53	W	Trial scanning from the west side
54	w.r.t.	abbreviation of “with respect to”
55	λ_n	Normalized eigenvalues
56	σ_r	Standard deviation reflectance of cluster points
57	Ψ	Geometric parameter

1. Introduction

Agricultural farms are required to increase production while reducing the environmental impact in a sustainable way (Tilman et al., 2011). Although mechanization and the evolution of agricultural machinery in extensive fields have helped to enhance crop production efficiency (Bechar and Vigneault, 2016), most of the intensive orchards are still being managed in similar ways to traditional farming methods, with labor intensive operations and without addressing in-field spatial

variability (Uribeetxebarria et al., 2019). To meet food supply and environmental demands, precision agriculture aims to find new strategies that allow the farmer for a more efficient management of orchards, reducing the amount of inputs while increasing fruit quality and productivity (Vázquez-Arellano et al., 2016).

To confront these challenges, orchard vigor and variability need to be better understood (Kamilaris and Prenafeta-Boldú, 2018). To this end, orchard characterization through information and communication technologies plays a crucial role, as shown in (Colaço et al., 2018a, 2018b; Narvaez et al., 2017). Obtaining an accurate characterization of trees by non-destructive methods at different growth stages provides valuable information that can be used for enhancing precision in orchard management (Andújar et al., 2017; Rosell and Sanz, 2012). This characterization can include phenology monitoring, plant geometric characterization and yield monitoring, among others.

In the last decade, different sensors and methods have been used for the geometric characterization of tree orchards. Due to the unstructured and complex nature of agricultural environments, with variable canopy structures in depth, porosity, training systems, among others (Vázquez-Arellano et al., 2016), the acquisition of 3D information –from depth cameras, structure-from-motion approaches, stereo vision and light detection and ranging (LiDAR) sensors– showed the most promising results in terms of orchard characterization and plant reconstruction, as has been shown in Rosell and Sanz (2012) and Yandún Narváez et al. (2016). Sensors and sensing techniques for the 3D modeling of orchards have been used to estimate parameters such as crop growth, height, shape and leaf area, with applications in pesticide treatments, irrigation, fertilization, pruning and crop training (Jiménez-Brenes et al., 2017; Narvaez et al., 2017; Pfeiffer et al., 2018; Sanz et al., 2018; Tagarakis et al., 2018).

For yield mapping and monitoring, the most commonly used sensors are RGB cameras (Gongal et al., 2015; Linker, 2017; Maldonado and Barbosa, 2016; Zhao et al., 2016). The main disadvantage of RGB cameras is that they only provide 2D information. Advances in artificial intelligence and computer vision have led to remarkable progress in fruit detection (Bargoti and Underwood, 2017a; Gan et al., 2018; Sa et al., 2016). Nevertheless, fruit detection performance continues to be affected by extrinsic factors that do not depend on the robustness of the algorithm, including the occlusion of fruits by other vegetative organs or lighting conditions (Bac et al., 2014; Gongal et al., 2015; Liu et al., 2016). Very few studies have tackled the problem of occlusions. Bulanon et al. (2009) and Gongal et al. (2018) proposed the use of multi-view imaging systems, while other works have used supportive tools such as an air blower to reduce melon leaf occlusions (Edan et al., 2000) or a mechanism to reduce canopy volume in citrus trees (Lee and Rosa, 2006). Other studies have considered variable lighting conditions, some of which have tried to minimize variable illumination effects by converting images to

other color spaces (Payne et al., 2013; Zhou et al., 2012), while others have proposed working with artificial lighting, although this involves the use of tunnel structures around trees or working at night time (Gongal et al., 2016; Linker and Kelman, 2015; Payne et al., 2014).

Other 2D sensors, including thermal, multispectral and hyperspectral cameras, have also shown potential for fruit detection (Bulanon et al., 2008; Okamoto and Lee, 2009; Sa et al., 2016; Safren et al., 2007; Zhang et al., 2015), although their use is not as extended as color cameras due to their cost and the high level of training required for their operation (Linker, 2018). The recent evolution in the fields of sensing and photonics has led to the introduction of the use of 3D sensors. Depth cameras based on stereoscopy, structured light or time-of-flight (ToF) are the most commonly used 3D sensors for fruit detection (Gené-Mola et al., 2019b; Gongal et al., 2015). The main limitation of these sensors is that their performance decreases under direct sunlight (Rosell-Polo et al., 2015).

LiDAR sensors have been widely applied for geometric characterization (Rosell and Sanz, 2012; Vázquez-arellano et al., 2016). However, their use is marginal for fruit detection and yield monitoring, probably because they are more expensive than other alternative sensors (such as RGB cameras) and do not provide color data. However, in addition to providing 3D information of the scene, one of the advantages of using LiDAR sensors is that the measurements are not affected by lighting conditions. Previous studies using LiDAR sensor data for fruit detection have been carried out in lab conditions and tested with a limited dataset, from 7 to 114 fruits (Feng et al., 2012; Gotou et al., 2003; Jiménez et al., 2000; Tanigaki et al., 2008). In other areas, LiDAR sensors have been placed on robotic platforms to infer the distance between the fruit and the robotic arm, though the actual detection of the fruit was performed using other systems such as RGB imaging (Bulanon and Kataoka, 2010; Ceres et al., 1998; Stein et al., 2016; Yin et al., 2009). More recently, the authors of the present study presented a proof of concept of the usefulness of LiDAR for apple detection in real commercial orchard environments (Gené-Mola et al., 2019a).

In this work, a multi-beam LiDAR sensor is used for remote fruit detection and plant geometrical characterization in a commercial Fuji apple orchard (*Malus domestica* Borkh. cv. Fuji). A fruit detection algorithm based on reflectance thresholding and Support Vector Machine (SVM) was developed. Under the hypothesis that moving the tree foliage and using multi-view sensing will reduce the number of fruit occlusions and will increase the percentage of fruits detected, the system was mounted on an air-assisted sprayer used to generate forced air flow. The rest of the paper is structured as follows: Section 2 presents the experimental setup, the fruit detection algorithm, and the methodology carried out to predict the yield and characterize the canopy; Section 3 evaluates the fruit detection algorithm and the effect of using air action and

different sensor positioning in terms of fruit detection accuracy, yield prediction estimation and geometric characterization performance; Section 4 discusses the performance of the system and compares the presented methodology to other works from the state of the art; finally, the conclusions retrieved from this work are presented in Section 5.

2. Methods

2.1. Experimental setup

Data was acquired in a commercial Fuji apple orchard (*Malus domestica* Borkh. cv. Fuji) (Fig. 1), located in Agramunt, Catalonia, Spain (E: 336297 m, N: 4623494 m, 312 m a.s.l., UTM 31T - ETRS89). The scanning was carried out 3 weeks before harvesting, at BBCH (Biologische Bundesanstalt, Bundessortenamt und CHemische Industrie) (Meier, 2001) growth stage 85. Trees grown in the selected orchard were 8-years-old and were trained in a tall spindle system with a maximum canopy height of 3.5-4 m, width of 1-1.5 m, and tree spacing of 4x1 m. All tests presented in this paper were carried out on 11 consecutive Fuji apple trees containing a total of 1444 apples (Table 1).

The equipment used for data acquisition was a mobile terrestrial laser scanner (MTLS) system with a multi-beam LiDAR sensor and a real-time kinematics global navigation satellite system (RTK-GNSS). A Puck VLP-16 (Velodyne LIDAR Inc., San José, CA, USA) LiDAR sensor was placed on a vertical plane to scan with a vertical field-of-view (FoV) of 360°, emitting 16 laser beams distributed in a horizontal FoV of 30°. In other words, each laser beam had a unique scanning angle, ranging from +15° to -15°, with a 2° step between the scanning angles. For each scan, the sensor provides a 3D point cloud with calibrated reflectance values (at 905 nm wavelength) of the measured scene, reporting values from 0-100 for diffuse reflectivities from 0% to 100% (Velodyne, 2016). This reflectance calibration was carried out by the sensor manufacturer, and allows getting reflectance values independently of laser power and distance. The LiDAR sensor acquisition frequency rate was set to 10 Hz (10 scans per second), corresponding to a vertical angular resolution of 0.2°. A GPS1200+ (Leica Geosystems AG, Heerbrugg, Switzerland) RTK-GNSS was used, with an absolute error of 0.01/0.02 m (horizontal / vertical), providing positioning measurements with rate of 20 Hz. Each sensor was connected to a rugged laptop, and data were synchronized by acquisition time stamp.

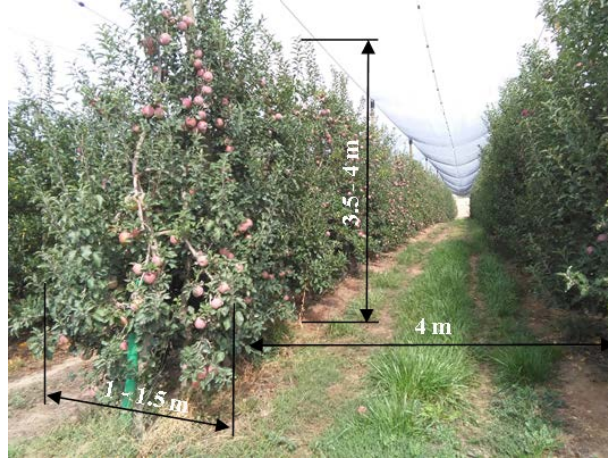


Fig. 1. Tested Fuji apple orchard.

The MTLs system was placed on an air-assisted sprayer and was pulled by a tractor at 0.125 m/s forward speed along a linear trajectory parallel to the row of trees. Since the MTLs system did not include an inertial measurement unit (IMU), moving the system at low speed (0.125 m/s) and along a linear trajectory was important to reduce vibrations (in amplitude) and obtain precise point clouds -without misalignments between different scans-. The air-assisted sprayer was used to generate turbulent air with the aim of moving the tree foliage and dis-occlude apples behind the leaves. As the LiDAR sensor was oriented vertically and the scanning plane was orthogonal to the canopy, the LiDAR measuring area was the area under the air flow influence (Fig. 2). The data acquired contains measurements from two different LiDAR heights (H_1 and H_2) and two different air conditions (n and af , where n stands for measurements without air flow action; and af for measurements with air flow action). The LiDAR sensor position H_1 corresponded to a sensor height of 1.8 m (approximately half of the tree height), while H_2 corresponded to the measurements with a LiDAR height of 2.5 m. In order to generate forced air, the air-assisted sprayer operated at $18\pi \text{ rad s}^{-1}$ (540 rpm of PTO, power take-off angular speed). In these conditions, the sprayer fan generated an air flow speed of $5.5 \pm 2.3 \text{ m s}^{-1}$, measured at a distance of 2.4 m from the fan axial center (the approximate horizontal distance between the sensor and the axis/trunk of the measured trees). A total of 4 trials were carried out, corresponding to all possible combinations between sensor heights and air conditions: $H_{1,n}$, $H_{1,af}$, $H_{2,n}$ and $H_{2,af}$. In order to obtain a complete representation of the canopy, for each trial, the tree row was scanned from the west (W) and from the east (E) side. Trials merging both scanned sides are denoted as (E+W).

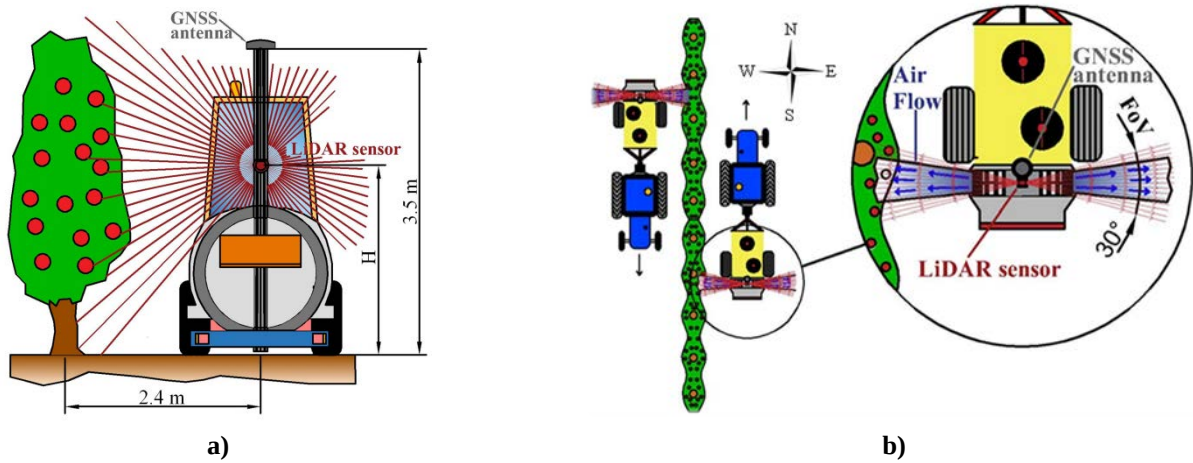


Fig. 2. Diagram of the MTLs and the arrangement of its elements. a) Rear view. b) Top view. The zoom-in circle shows the position of the LiDAR sensor with respect to the sprayer fan. Air flow is represented in blue, while red lines illustrate the 16 laser beams distributed along the sensor field-of-view (FoV).

For each scan, a 3D point cloud with coordinates relative to the sensor was provided by the LiDAR system. To generate the point clouds with absolute coordinates, the RTK-GNSS data were used to infer the position and the orientation of the sensor, obtaining a rotation and translation matrix that transformed points in relative coordinates to points in absolute global world coordinates. To generate ground truth of the apples locations (GT_{labels}), the resulting point clouds were manually labelled, placing 3D rectangular bounding boxes around each apple, as shown in Fig. 3c. This annotation was carried out using the software CloudCompare (Cloud Compare [GPL software] v2.9 Omnia) and supported by additional RGB images of the tested trees. A total of 1353 fruits were visually identified in the point cloud during ground truth generation, representing the 93.7% of the total amount of fruits manually counted (GT_{field}) in the orchard. Thus, 6.3% of apples were discarded since they were not visible in the 3D point clouds. Table 1 shows the number of fruits per tree manually counted in the orchard GT_{field} compared with the number of labelled apples in the 3D point cloud GT_{labels} . The dataset generated and analyzed during the current study has been made publicly available at http://www.grap.udl.cat/en/publications/LFuji_air_dataset.html.

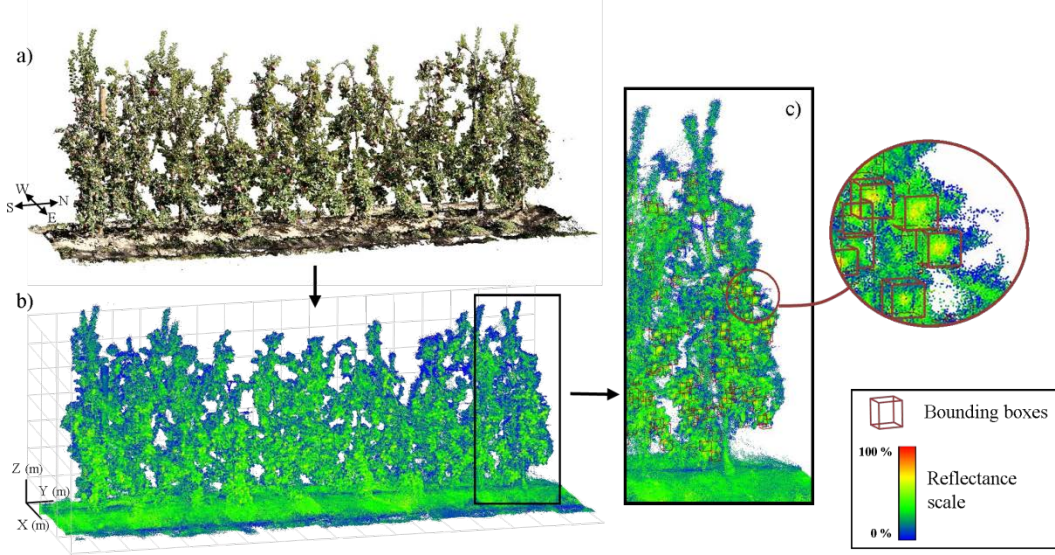


Fig. 3. Illustration of the dataset generated for the current study. a) RGB image of the measured Fuji apple trees. b) Point cloud data generated from MTLs measurements. Color scale ranges from 0% (blue) to 100% (red) corresponding to the calibrated reflectance of the measured scene. c) Annotated point cloud with 3D rectangular bounding boxes placed around each apple.

Table 1. Fruit counting ground truth. Comparison between the number of fruits manually counted (GT_{field}) in the orchard and the number of fruits annotated in the 3D point cloud (GT_{labels}).

	GT_{field}	GT_{labels}
Tree 01	139	138
Tree 02	106	100
Tree 03	139	131
Tree 04	137	129
Tree 05	94	85
Tree 06	131	119
Tree 07	119	114
Tree 08	145	137
Tree 09	139	131
Tree 10	136	122
Tree 11	159	147
Total	1444	1353

2.2. Fruit detection algorithm

With the purpose of detecting and locating fruits from the MTLs data, the algorithm presented in Gené-Mola et al. (2019a) was implemented but lightly modified. The main modification included the use of an SVM approach (Cortes and Vapnik, 1995) in order to avoid using manually set parameters and to automatically train the features that characterize apples. The algorithm consists of 4 main steps: pre-processing, fruit clustering, fruit separation and false positive removal

(Fig. 4). To combine different scans, such as data from east and west sides, from different sensor heights or different air flow conditions, the 3D point clouds were merged before applying the fruit detection algorithm. The registration of different point clouds was automatic since all scans were georeferenced in absolute world coordinates. Merging different scans in a single point cloud allows for reducing the number of multi-detections, e.g., points from an apple appearing in two different scans (such as from east and west sides) were merged in a unique point cloud, and in consequence, the apple was detected only once.

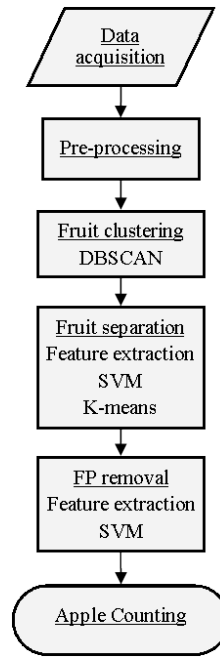


Fig. 4. Fruit detection algorithm pipeline.

Pre-processing was based on the fact that apples have a higher IR reflectance than background. In this step, a reflectance threshold was set in order to remove the points that are not likely to belong to an apple. Then, sparse outlier removal (Rusu et al., 2008) was applied to remove noisy points. After removing background points, the remaining points were clustered in groups of connected points by applying a density-based scan algorithm DBSCAN (Ester et al., 1996). These first two steps (pre-processing and fruit clustering) were implemented following the algorithm presented in Gené-Mola et al. (2019a).

The minimum distance used in DBSCAN to cluster connected points was $\varepsilon = 0.03$ m. Therefore, points belonging to two different fruits closer than ε were grouped in the same cluster. A fruit separation step was applied to discriminate the aforementioned clusters. First, the features of each cluster (volume, number of points, eigenvalues, and reflectance) were extracted. Then, a linear SVM with a penalty factor of $C=0.35$ (Burges, 1998) was used to predict the number of fruits (K)

that contains each cluster. Clusters that had more than one apple ($K > 1$) were split into K sub-clusters using the K-means algorithm (Jain, 2010).

The last step of the algorithm was a false positive filter. False positives are the detections that have been wrongly classified as apple. These false positives were derived from elements –such as trunks or leaves– that had reflectance values higher than expected ($R > 60\%$), or from clusters that were wrongly split, detecting apples more than once (multi-detections). An SVM was used to classify each cluster as a correct or wrong detection. The SVMs used in the fruit separation and false positive removal steps were fed with 8 cluster features:

- Cluster volume V .
- Number of points Pt that contain a cluster.
- Normalized eigenvalues $\lambda_n = [\lambda_{1n}, \lambda_{2n}, \lambda_{3n}]$. The eigenvalues are obtained by singular value decomposition (SVD), and their value depends on the variance of points (3D data) projected on their principal axes (Jolliffe, 2011). In order to compare different clusters, an eigenvalues normalization is applied, so that the sum of normalized eigenvalues is one.
- Geometrical parameter $\Psi = 27 \cdot \lambda_{1n} \cdot \lambda_{2n} \cdot \lambda_{3n}$ defined as the product of normalized eigenvalues and a normalization factor equal to 27. Since the maximum value of the product of the normalized eigenvalues is equal to $\frac{1}{27}$, achieved when all normalized eigenvalues are equal to $\frac{1}{3}$ (for spherical clusters), the normalization factor of 27 allows the geometrical parameter to be bounded between 0 and 1, being 1 for spherical clusters.
- Reflectance histogram $rh = [rh_1, rh_2, rh_3, rh_4, rh_5] / P$, where $rh_1, rh_2, rh_3, rh_4, rh_5$ are the number of points in the cluster with a reflectance between 60:68, 68:76, 76:84, 84:92 and 92:100, respectively.
- Mean reflectance of cluster points $\bar{R}$.
- Standard deviation reflectance of cluster points σ_R .
- Maximum reflectance of cluster points R_{max} .

All processing presented in this work was implemented using MATLAB[®] (R2018a, Math Works Inc., Natick, Massachusetts, USA) and is publicly available jointly with the corresponding dataset at http://www.grap.udl.cat/en/publications/LFuji_air_dataset.html.

2.3. Canopy characterization

The mean canopy height, width, contour and cross-section area were computed following the methodology described in Escolà et al. (2017). The row of trees was splitted into vertical slices of 0.1 m length. For each slice, the maximum canopy

height and width were computed as the distance between the two most distant points in the vertical and horizontal directions (denoted as Z and X axis in Fig. 3b, respectively). Then, the mean height and mean width were calculated as the mean of the maximum canopy heights and maximum canopy widths of all slices (110 slices for all 11 trees). The mean canopy contour was obtained similarly, computing the mean canopy width at different tree heights intervals of 0.1 m (40 heights intervals). The area within the mean canopy contour was defined as the cross-section area.

Finally, the leaf area was estimated using the projected tree row surface (PTRS) described in Sanz et al. (2018), which correlates the frontal projected surface and the top projected surface with the linear leaf area (leaf area per meter).

2.4. Performance evaluation

The effect of using forced air action and different sensor positions (multi-view sensing) was evaluated in terms of fruit detection accuracy, yield prediction estimation and geometric characterization performance.

To evaluate the fruit detection accuracy, each detection obtained using the fruit detection algorithm was classified as one of the following groups:

- True positive (TP): Detections that intersect with a ground truth apple label (bounding box annotation) with an overlap higher than 50%. In case of multi-detections, only one true positive was counted.
- False positive (FP): Detections that do not intersect with an annotation with an overlapping higher than 50%.
- Multi-detection (MD): A multi-detection is produced when a single apple is detected n times (by different detections). That could happen, for instance, if a single apple detection was wrongly split in K detections when applying the fruit separation step. In that case, it is counted one TP and $n - 1$ multi-detections MD .

Having the total amount of TP , FP , MD , and the number of labels detected (LD), the fruit detection accuracy is assessed in terms of detection rate (DR), recall (R), precision (P), false positive rate (FPR), multi-detection rate (MDR) and F_1 -score, as follows:

$$DR = \frac{LD}{T}, \quad (1)$$

$$R = \frac{TP}{T}, \quad (2)$$

$$P = \frac{TP}{D}, \quad (3)$$

$$FPR = \frac{FP}{D}, \quad (4)$$

$$MDR = \frac{MD}{D}, \quad (5)$$

$$F_1 = 2 \frac{R \cdot P}{R + P}, \quad (6)$$

where D is the number of fruit detections and T is the total number of fruits in the dataset.

Since the algorithm used for fruit detection needs to be trained, the test has to be performed using a new dataset, different to the one used for training. To do so, an 11-fold cross-validation was practiced (11 iterations). Each iteration evaluated one tree, while the other trees were used as training set. The final test results presented in sections 3.1 and 3.2 were obtained aggregating TP , FP and MD from all iterations and computing the metrics previously defined for all the dataset. Section 3.1 reports results after the pre-processing (Pp), fruit separation (FS) and false positive removal (FPr) steps. Different combinations of features used in the FS and FPr steps were evaluated in order to assess the usefulness of using each feature. This evaluation was carried out using data acquired from sensor height H_1 , without forced air action and from both row sides ($H_{1,n,(E+W)}$). On the other hand, section 3.2 evaluates, both qualitatively and quantitatively, the fruit detection performance under different conditions and analyzes the effect of using forced air flow and scanning at different sensor heights. Results are reported either with respect to (w.r.t.) GT_{field} , as well as w.r.t. GT_{labels} . These two different approaches allow comparing the present methodology to other fruit detection works evaluated w.r.t. the number of visible fruits (Gongal et al., 2015).

The yield prediction was also evaluated following the 11-fold cross-validation model. First, the fruit detection algorithm was used to automatically count the number of fruits on each tree. Then, 11 iterations were performed in order to assess the yield prediction on each tree. In yield prediction, what is important is not so much the percentage of fruits detected but rather the correlation that exists between the number of detections and the actual number of fruits in the tree (Linker, 2017). Thus, a simple linear regression model $y = a \cdot x + b$ was obtained using the training set. This model related the number of fruits detected D and the actual number of fruits manually counted in the field GT_{field} . Then, the linear model was used to predict the number of fruits of the test set. With that, we ensured that the prediction model was not influenced by the detections of the tested tree. Finally, the prediction error associated to each tree prediction was computed as

$$Error_t = \frac{FruitsPredicted - GT_{field}}{GT_{field}}. \quad (7)$$

In order to have an evaluation of all the dataset, the root mean square error (RMSE) was computed:

$$RMSE = \sqrt{\frac{\sum_{t=1}^N (Error_t)^2}{N}}, \quad (8)$$

where N is the number of trees evaluated; in this work $N=11$.

For the geometrical characterization evaluation, the trial $H_{1,n}$ was considered the reference measurement as it was the configuration validated in the original methodologies used to compute the geometrical parameters assessed (Escolà et al., 2017; Sanz et al., 2018). Since training data were not required, this evaluation was carried out on all the dataset at once, comparing the geometrical characterization obtained with standard scanning, $H_{1,n}$, with forced air flow application, $H_{1,(n+af)}$, and with the multi-view approach $H_{(1+2),n}$.

3. Results

This section evaluates the fruit detection algorithm and analyses the effect of air flow and different sensor positioning for fruit detection, yield prediction and geometric characterization. All results are obtained using the data presented in section 2.1.

3.1. Feature assessment

Comparing results using all features, the FS and FPr steps significantly improved the algorithm performance, going from an F_1 -score of 0.7449 (after pre-processing) to an F_1 -score of 0.7837 and 0.8119 after the FS and FPr steps, respectively (Table 2). Regarding the features, volume was the most useful for FS, presenting an F_1 -score improvement of 4% (from 0.7449 to 0.7824), however, it did not contribute in removing FP. The most useful feature for FPr was the reflectance histogram, improving the F_1 -score by 2% when it was used. It can be observed that the other tested features also contributed positively to the FS and FPr steps, except for σ_R and R_{max} which slightly penalized the FS although they were useful for FPr.

Table 2. Features assessment using data acquired at sensor height H1 without forced air action ($H_{1,n}$). The evaluation is reported in terms of F_1 -score with respect to the number of annotated fruits.

Feature	Pp									
V		✓	✓	✓	✓	✓	✓	✓	✓	✓
rh			✓	✓	✓	✓	✓	✓	✓	✓
ψ				✓	✓	✓	✓	✓	✓	✓
Pt					✓	✓	✓	✓	✓	✓
λ_n						✓	✓	✓	✓	✓
$\bar{R}$							✓	✓	✓	✓
σ_R								✓	✓	✓
R_{max}										✓
Pp + FS	0.7449	0.7824	0.7834	0.7838	0.7842	0.7843	0.7847	0.7839	0.7837	
Pp + FS + FPr	0.7449	0.7824	0.8019	0.8089	0.8090	0.8098	0.8103	0.8110	0.8119	

3.2. Fruit detection results

For simplicity, in this section results w.r.t. GT_{labels} are presented outside parentheses, together with results w.r.t. GT_{field} inside parentheses. Trials with data from only one tree side (either E or W) presented the lowest detection rates, reporting

F₁-score values of 0.537 (0.513) and 0.624 (0.598), at conditions $H_{1,n,E}$ and $H_{1,n,W}$, respectively. The performance significantly improved when the trees were scanned from both sides, presenting an F₁-score of 0.812 (0.784) at conditions $H_{1,n,(E+W)}$. The detection rates dropped when trees were scanned under forced air flow conditions $H_{1,af,(E+W)}$ because the point cloud quality decreased (became “blurred”) and, while some apples were dis-occluded by the air flow effect, others were occluded (Fig. 5). The forced air flow usefulness was found when combining the data acquired with and without forced air flow $H_{1,(n+af),(E+W)}$. Merging these trials, the percentage of detected fruits increased by more than 7%, achieving a detection rate of 0.894 (0.838), a recall of 0.814 (0.763) and an F₁-score of 0.826 (0.799). Trials at height H_2 presented lower detection rates because the LiDAR was positioned in an upper position and, in consequence, the system failed on detecting fruits at the bottom parts of trees (Fig. 6 and Fig. 8). Nevertheless, combining trials at different sensor heights $H_{(1+2),n,(E+W)}$ (multi-view approach) increased fruit detection performance similarly to combining different air conditions, achieving an F₁-score of 0.830 (0.802). Finally, combining all trials $H_{(1+2),(n+af),(E+W)}$ (different sensor heights and air flow conditions) increased the detection rate but penalized the recall. This took place due to fact that the point cloud became more blurred, making difficult to split up groups of apples.

Table 3. Fruit detection assessment at different sensor heights and air flow conditions. Results are reported in terms of detection rate (DR), recall (R), precision (P), false detection rate (FDR), multi-detection rate (MDR) and F₁-score. DR, R and F₁-score were computed with respect to the number of labelled fruits (L), and to the total amount of fruits manually counted in the field (F). Best achieved results are in bold type.

Trial	DR		R		P	FDR	MDR	F ₁ -score	
	L	F	L	F				L	F
*$H_{1,n,(E+W)}$	0.823	0.771	0.758	0.710	0.875	0.104	0.021	0.812	0.784
$H_{1,n,E}$	0.415	0.389	0.383	0.359	0.898	0.089	0.013	0.537	0.513
$H_{1,n,W}$	0.540	0.506	0.485	0.454	0.875	0.110	0.015	0.624	0.598
$H_{1,af,(E+W)}$	0.768	0.720	0.698	0.654	0.868	0.108	0.024	0.774	0.746
$H_{1,(n+af),(E+W)}$	0.894	0.838	0.814	0.763	0.839	0.110	0.051	0.826	0.799
$H_{2,n,(E+W)}$	0.663	0.621	0.588	0.551	0.841	0.131	0.028	0.692	0.666
$H_{2,n,E}$	0.351	0.329	0.318	0.298	0.909	0.085	0.006	0.471	0.449
$H_{2,n,W}$	0.429	0.402	0.369	0.346	0.798	0.171	0.031	0.505	0.482
$H_{2,af,(E+W)}$	0.573	0.537	0.517	0.484	0.885	0.096	0.019	0.652	0.626
$H_{2,(n+af),(E+W)}$	0.748	0.701	0.656	0.615	0.803	0.143	0.054	0.722	0.696
$H_{(1+2),n,(E+W)}$	0.892	0.836	0.802	0.751	0.860	0.095	0.045	0.830	0.802
$H_{(1+2),n,E}$	0.528	0.495	0.480	0.450	0.884	0.096	0.020	0.622	0.596
$H_{(1+2),n,W}$	0.653	0.612	0.576	0.540	0.853	0.112	0.035	0.688	0.661
$H_{(1+2),af,(E+W)}$	0.868	0.813	0.793	0.743	0.866	0.092	0.042	0.828	0.800
$H_{(1+2),(n+af),(E+W)}$	0.917	0.859	0.789	0.739	0.777	0.101	0.122	0.783	0.758

315 For a visual/qualitative assessment, Fig. 5 shows an example of the acquired point cloud and the corresponding fruit
316 detection results. Due to the field-of-view of the RGB camera used to acquire images shown in Fig. 5a, only the bottom
317 part (from 0m to 2m height) of tree 4 was illustrated. To help the visualization, the colour scale of Fig. 5b represents the
318 calibrated reflectance of the measured scene, illustrating reflectance values from 10% (blue) to 80% (red). From that, it can
319 be observed that apples present a higher reflectance than other tree elements.

320 Red squares of Fig. 5 show an example of an apple that was occluded in trial $H_{1,n}$, and became dis-occluded in $H_{1,af}$,
321 due to the effect of applying forced air flow. The opposite happened in the blue squares example, where an apple that was
322 visible in $H_{1,n}$ became occluded in $H_{1,af}$. These occlusions affected the fruit detection performance, as it can be seen in the
323 detections shown in Fig. 5c. To take advantage of the dis-occlusions produced by the air flow effect without being
324 penalized by its occlusions, the data from both trials ($H_{1,n}$ and $H_{1,af}$) were combined, increasing the number of visible apples
325 as it can be observed in the detections shown in $H_{1,(n+af)}$. Comparing the detection rates between $H_{1,n,(E+W)}$ and $H_{1,(n+af),(E+W)}$
326 in Table 3, more than a 7% of fruits were dis-occluded when combining trials with and without force air flow, presenting
327 an increase of the DR from 0.823 in $H_{1,n,(E+W)}$ to 0.894 in $H_{1,(n+af),(E+W)}$.

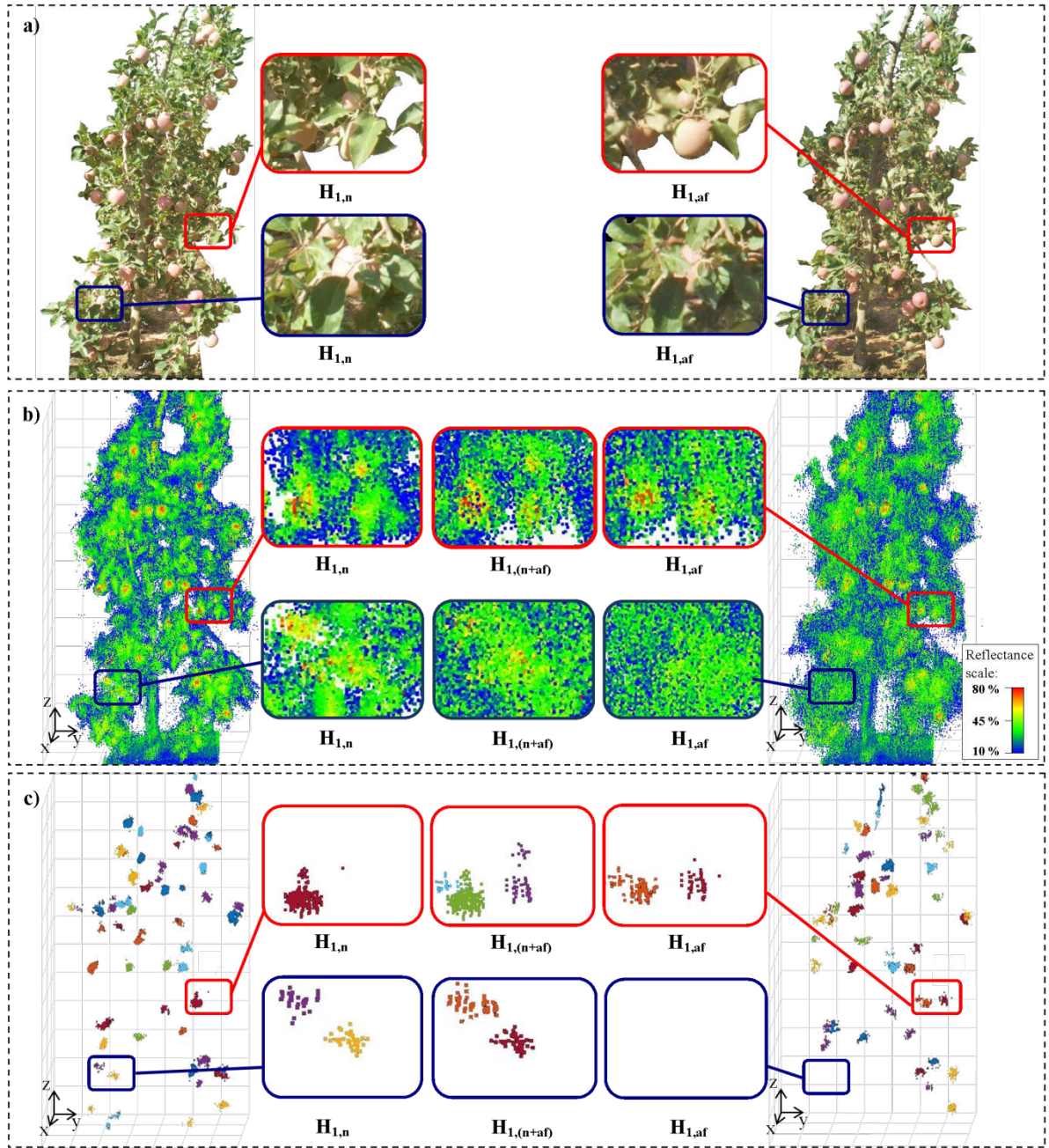


Fig. 5. Illustration of the forced air flow effect in fruit detection. Left side corresponds to a trial without forced air ($H_{1,n}$) while right side corresponds to a measurement with forced air flow action ($H_{1,af}$). a) RGB images taken from the sensor position. b) Point cloud obtained with the MTLs. c) Fruit detections using the fruit detection algorithm. Squares in red and blue are a zoom-in of two zones where the air flow effect dis-occluded (red) or occluded (blue) some fruits. Squares denoted by $H_{1,(n+af)}$ correspond to the trial that aggregates the data from $H_{1,n}$ and $H_{1,af}$.

3.3. Fruit location results

As well as not being affected by usual field lighting conditions, the LiDAR sensor used has the advantage of providing the relative 3D location of the fruits detected. When integrated in the designed MTLs, the fruits are located in global 3D

coordinates. That allows to obtain the spatial distribution of detected fruits in height (along 'y' axis) and in depth (along 'x' axis). Fig. 6 shows that that most of the fruits from the dataset (11 trees) were between heights of 0.6 m and 2.1 m. It is also observed that, when scanning at height H_2 , the detection rate dropped in lower zones of the tree, while aggregating data from both scanning heights $H_{(1+2)}$ helps to increase the detection rate (Fig. 6). The distribution of fruits along the 'x' axis (Fig. 7) shows that the east side ($x > 0$) was 10cm wider and had a 2 % more of fruits than the west side ($x < 0$). Due to the limited number of tested trees (11 trees), further tests should be carried out to study the relation between the canopy width and the fruit yield. The fruit distribution also shows that 1233 fruits out of 1353 (91%) were at distances between -0.4 m and 0.4 m from the center of the tree (along the x axis), with a maximum production of 599 (44%) fruits at distance intervals of $\pm[0.1, 0.3]$ (Fig. 7). It is also observed that the best detection performance was achieved when combining data acquired with and without forced air action (cyan vertical bars). It should be noted that the mean canopy contour was computed from the average of all canopy widths in the data set and not as the maximum width. For this reason, some detected apples in Fig. 7 fall outside the plotted mean canopy contour.

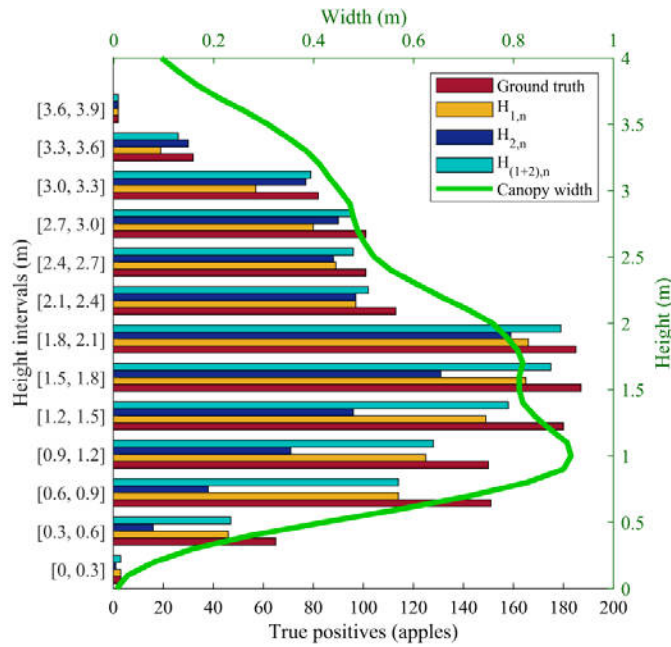


Fig. 6. Distribution of fruits in height and comparison with the number of fruits identified when scanning at different sensor height (H_1 and H_2). Data includes information from all dataset (11 trees). Left and bottom axis refers to the horizontal bars, which provide the number of true positives identified by the fruit detection algorithm at different height intervals. Right and top axis (height and width) refers to the mean canopy width illustrated in green.

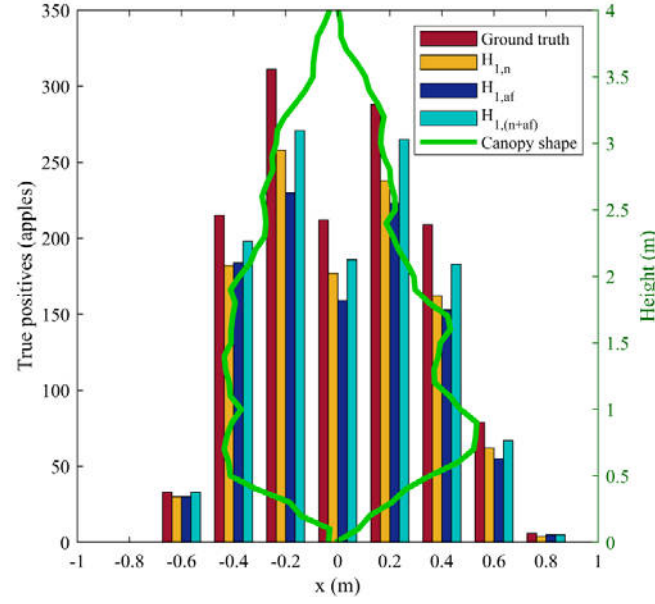


Fig. 7. Distribution of fruits in depth (along x axis) and comparison with the number of fruits identified in all dataset (11 trees) when scanning at different air flow conditions (n and af). The mean canopy contour is also illustrated in green. Left and bottom axis refers to the number of true positive identified in each ' x ' position, while the right axis refers to the height of the mean canopy contour.

From the distribution histograms presented in Fig. 6 and Fig. 7, it is straightforward to obtain an evaluation of the fruit detection performance at different tree locations, facilitating an analysis of where the fruit detection system fails or succeeds. Fig. 8 illustrates the detection rate (DR) distribution in height and along the x axis for the different trials: height sensor positions H_1 and H_2 , air flow conditions n and af , and scanned sides E and W . Since in H_1 the sensor height was approximately the half of the tree height, the DR of trial H_1 decreased in the lower and upper tree zones (more sharply in the upper zone), approximately at heights under 1m and above 2.5m (Fig. 8a). On the other hand, in H_2 the sensor was located in the upper zone, which explains that the DR of trial H_2 decreased in the lower parts of the tree, under 2m height, while reporting DR > 85% for heights above 2.5m (Fig. 8a). The trial that combines both sensor heights $H_{(1+2)}$ takes advantage of both views, presenting DR higher than 85% for tree heights above 1m. Regarding the detection performance in depth (along the ' x ' axis), H_2 presented low DR in external zones -further than 0.4m from the center of trees- (Fig. 8b). This is because the widest zones were at the bottom of the tree, corresponding to the zone where H_2 had more occlusions. In Fig. 8(c-f), the DR was higher in the trials without forced air flow than in the trials with forced air flow. Nevertheless, the DR was improved more than 5% when combining both scanning conditions. This improvement was seen along all tree heights and widths. Finally, Fig. 8(g-h) illustrate the DR distribution when processing data from tree row sides separately (E or W) and together ($E+W$). The detection rate dropped more than 28% on the non-scanned side: trial $H_{1,n,E}$ presented low detection rates on the west side ($x < 0$) and trial $H_{1,n,W}$ on the east side ($x > 0$). However, when aggregating the two sides, the detection performance was almost constant in all tree widths because of the reduction of the number of fruit occlusions when scanning from the two sides of the apple trees.

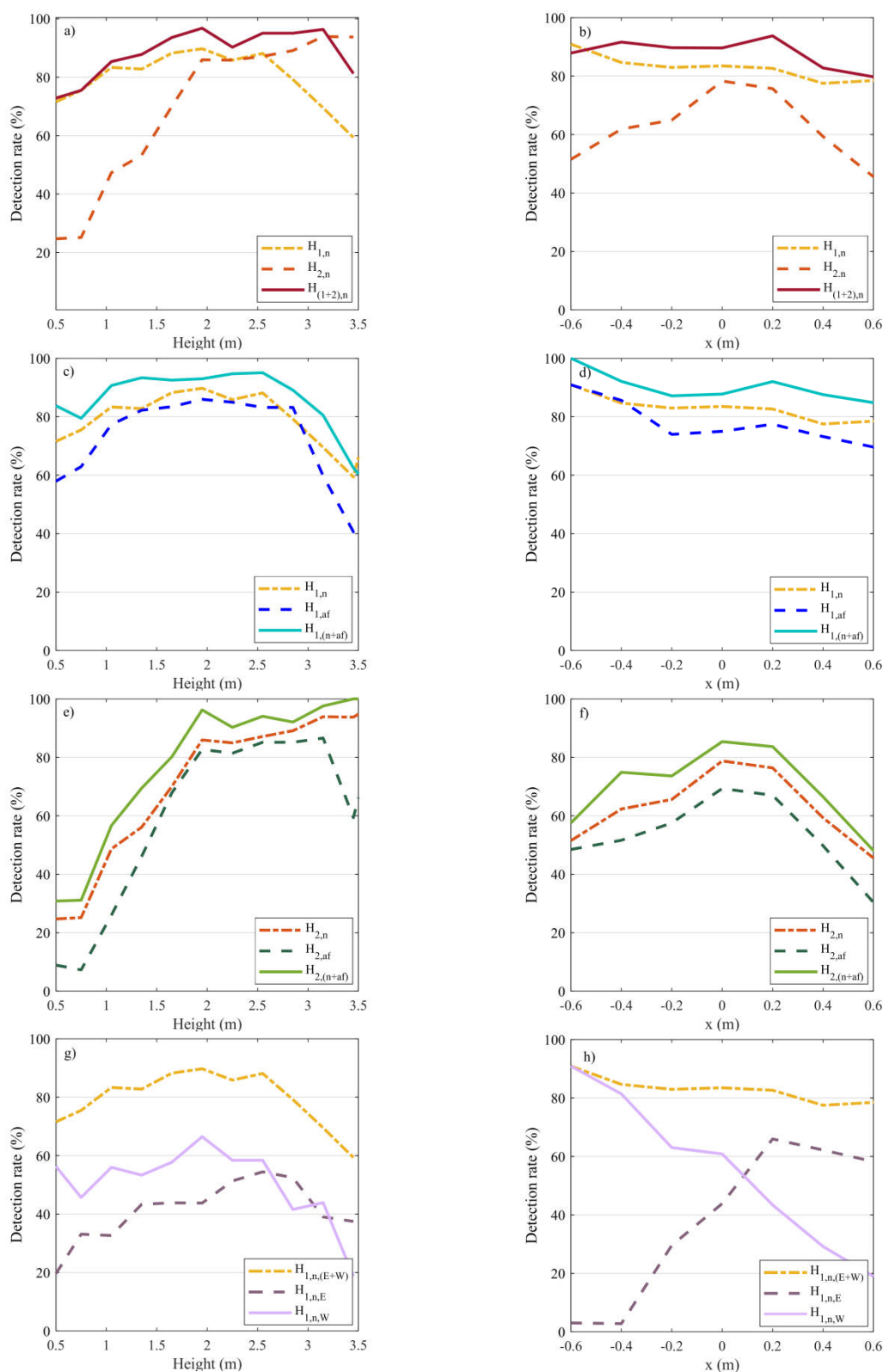


Fig. 8. Detection rate under different air flow conditions (n and af) and sensor positions (H_1 and H_2) at different tree locations: in height (a, c, e, g) and along the 'x' axis (b, d, f, h). All plots evaluate data acquired from both row sides (E+W), except g) and h) where the evaluated row side is specified in the corresponding legend.

3.4. Yield prediction results

Table 4 shows yield prediction results for trial $H_{(1+2),n}$, which reported the highest F_1 -score, as seen in section 3.2. Results showed higher prediction errors when using data from only one tree side (E or W), obtaining RMSEs of 19.0% and 12.4% and a coefficient of determination (R^2) of 0.58 and 0.54, when scanning from the east and west sides, respectively. The prediction was significantly improved when using data from both tree sides ($E+W$), presenting errors between -7.7% and 12.0% with an RMSE of 5.7% (Table 4) and $R^2=0.87$ (Fig. 9).

Table 4. Yield predictions results (number of fruits) combining H_1 and H_2 data without forced air flow ($H_{(1+2),n}$). Results are presented for data acquired from one tree side, either east (E) or west (W), and combining data from both sides ($E+W$).

	Ground Truth (# fruits)	Fruits detected (D)			Fruits predicted (a·D + b)			Prediction error (%)		
		E	W	$E+W$	E	W	$E+W$	E	W	$E+W$
Tree01	139	71	96	125	133.9	145.7	142.8	-3.7	4.8	2.7
Tree02	106	57	69	98	126.6	119.9	115.5	19.5	13.1	9.0
Tree03	139	64	81	118	128.3	128.2	134.4	-7.7	-7.7	-3.3
Tree04	137	61	89	116	126.0	137.4	132.3	-8.0	0.3	-3.4
Tree05	94	44	59	79	121.0	116.2	92.8	28.8	23.7	-1.3
Tree06	131	66	96	127	130.7	147.5	146.7	-0.2	12.6	12.0
Tree07	119	54	84	103	122.5	133.6	118.9	3.0	12.3	-0.1
Tree08	145	66	77	118	129.3	122.7	133.8	-10.8	-15.4	-7.7
Tree09	139	63	90	126	127.5	138.3	144.0	-8.3	-0.5	3.6
Tree10	136	68	70	114	131.8	114.0	130.1	-3.1	-16.2	-4.4
Tree11	159	120	102	137	237.3	147.0	153.4	49.3	-7.6	-3.5
Total	1444	734	913	1261	RMSE:			19.0	12.4	5.7

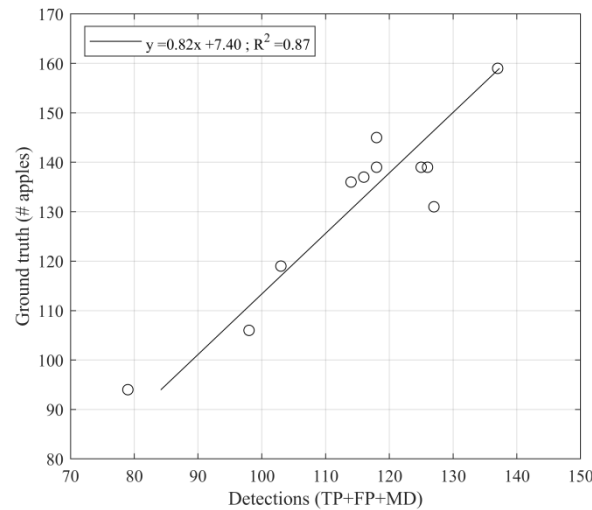


Fig. 9. Linear regression between the number of apples detected with $H_{(1+2),n,(E+W)}$ and the actual number of apples per tree (GT_{field}).

When comparing different sensor heights and air flow conditions, the highest performance was achieved by $H_{1,n,(E+W)}$, obtaining an RMSE of 5.4% (Table 5). Although merging different air conditions ($n+af$) or different sensor heights ($H_{(1+2)}$) improved the percentage of fruits detected, neither the air flow effect $H_{1,(n+af),(E+W)}$ nor the multi-view approach $H_{(1+2),n,(E+W)}$ improved yield prediction, presenting similar results to those of trial $H_{1,n,(E+W)}$.

Table 5. Yield prediction assessment at different sensor heights ($H1$ and $H2$), air conditions (n and af) and scanned sides (E and W). Best achieved results are in bold type.

Trial	RMSE (%)
$H_{1,n,(E+W)}$	5.4
$H_{1,n,E}$	15.2
$H_{1,n,W}$	15.3
$H_{1,af,(E+W)}$	6.8
$H_{1,(n+af),(E+W)}$	5.5
$H_{2,n,(E+W)}$	8.1
$H_{2,n,E}$	11.3
$H_{2,n,W}$	10.6
$H_{2,af,(E+W)}$	12.7
$H_{2,(n+af),(E+W)}$	10.0
$*H_{(1+2),n,(E+W)}$	5.7
$*H_{(1+2),n,E}$	19.0
$*H_{(1+2),n,W}$	12.4
$H_{(1+2),af,(E+W)}$	6.7
$H_{(1+2),(n+af),(E+W)}$	8.1

3.5. Geometric characterization results

Regarding height results, it was observed that the forced air flow and multi-view approaches produced very similar results when measuring the mean canopy height. Mean width estimation was neither significantly affected by the multi-view configuration; but a difference higher than 10% was reported when scanning with forced air flow (Table 6). The mean cross-section area measurement was the parameter which has been affected the most by the scanning conditions, with differences of 22.3% when combining air conditions and 16.3% when combining different sensor heights. These higher deviations are the consequence of an error propagation of height and width measurements. For a qualitative evaluation, Fig. 10 illustrates the mean canopy contour obtained in different trials. It can be observed that the multi-view approach (plotted in blue) matched slightly better the reference trial than the trial that combines different air conditions (plotted in red). Finally, concerning the leaf area analysis, both approaches performed similarly, with a 10.6% difference when combining different air conditions and an 8.3% difference with the multi-view approach.

Table 6. Geometric characterization assessment at different sensor heights ($H1$ and $H2$) and air conditions (n and af). Differences with respect to the reference trial $H1_n$ are reported within brackets.

Height	Air flow	Mean Height [m]	Mean Width [m]	Mean Cross-Section area [m ²]	Mean Leaf Area [m ² /m]
H1	n	3.64	1.23	2.12	9.77
H1	(n+af)	3.71 (2.1%)	1.36 (10.7%)	2.59 (22.3%)	10.80 (10.6%)
(H1+H2)	n	3.68 (1.3%)	1.27 (3.8%)	2.46 (16.3%)	10.57 (8.3%)

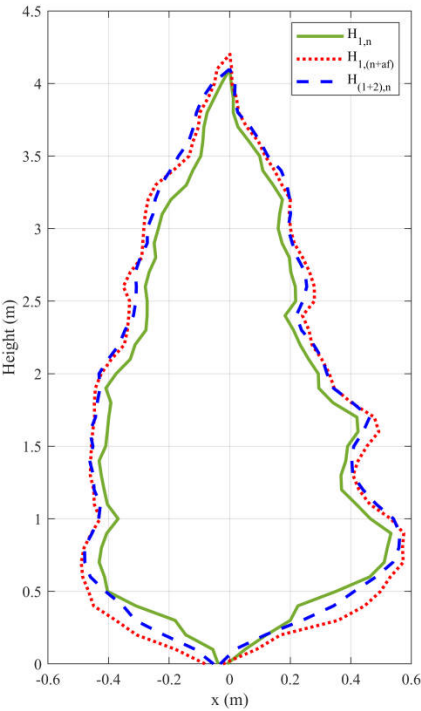


Fig. 10. Illustration of the mean canopy contour obtained at different sensor heights ($H1$ and $H2$) and air flow conditions (n and af). The east side (E) corresponds to the positive horizontal distances.

4. Discussion

The fruit detection results showed that trials $H_{1,(n+af)}$, and $H_{(1+2),n}$, located 6.7 % and 6.5 % more fruits than $H_{1,n}$, respectively, presenting $DR > 0.89$ and $R > 0.80$ with respect to the total number of annotated fruits. Although it is difficult to compare systems tested with different datasets, these results are similar (in terms of detection rate) to those obtained in studies based on color images, which have reported accuracies of between 80% and 85% using color features (Gongal et al., 2015) and up to 90% (F_1 -score) using deep learning (Bargoti and Underwood, 2017a; Gené-Mola et al., 2019c; Sa et al.,

2016). The results are also comparable with those of studies based on vision systems used in orchard harvesting robots, which have reported a mean location success of 80% and a mean identification success of 70% (Bac et al., 2014).

The system used in this study was based on the MTLs described in Escolà et al. (2017). However, there was a big improvement when replacing the 2D LiDAR sensor with the 3D Velodyne VLP-16. Although the original system was not used to detect fruits, having 16 laser beams within a $\pm 15^\circ$ horizontal FoV, that is 16 different points of view, certainly contributed to improving the fruit detection rates. In future works, the system could be further improved by including an IMU sensor. With that, the system could scan faster -higher forward speed- and along different types of trajectories -not limited to linear-.

Among the main advantages of using the system presented here for fruit detection are that the sensor is not affected by lighting conditions and provides the 3D location of fruits. This allows to know the special fruit distribution in the tree, which can be useful for studying and optimizing agricultural processes (Martin-Gorriz et al., 2014; Widmer and Krebs, 2001), e.g. comparing different pruning and thinning strategies with the spatial distribution of fruits in the trees and the yield. In Fig. 6, it was observed that in general terms the number of fruits increases in wider zones, and in Fig. 7 that the number of fruits at the center of the tree was 30% lower than at horizontal distance intervals of $\pm[0.1, 0.3]$. This could be a consequence of canopy porosity (Pfeiffer et al., 2018; Trentacoste et al., 2018), because the center of the tree receives less light than the outer parts, however further analysis is needed. System performance at different heights and widths was analyzed. Fig. 8a showed that the multi-view approach $H_{(1+2),n}$ presented similar detection rates at different heights. In contrast, with $H_{1,n}$ system performance decreased when trying to detect fruits from the upper zone of the tree, and with $H_{2,n}$ when detecting fruits from the bottom. As for fruit detection performance along the horizontal axis, $H_{2,n}$ presented low detection rates in the outer parts, whereas $H_{1,n}$ and $H_{(1+2),n}$ presented similar detection rates in all tree widths (Fig. 8b).

Knowing the fruit distribution on the tree structure could be valuable for the planning and optimization of harvesting strategies (Bargoti and Underwood, 2017b). For example, depending on the amount of fruits in the top parts of the trees and considering the extra costs involved to pick them (use of ladders or elevation platforms), the farmer could decide not harvest the highest areas. This could also result in improved overall quality of the harvested apples, due to the reduction of fruits damage (in lower and intermediate parts of the tree), caused by metal structures needed to reach the highest points (Młotek et al., 2015).

With respect to the yield prediction results, the presented system was able to estimate the number of fruits on each tree with an RMSE of 5.4% when scanning with standard conditions $H_{1,n}$. Similar results were obtained using the forced air

flow and multi-view approaches, with RMSE values of 5.5% and 5.7%, respectively. This means that, although merging air conditions or sensor heights can help to minimize the number of fruit occlusions, these conditions do not provide an advantage for yield prediction, because the correlation between the number of detections and the actual number of fruits in the trees was similar to the one in $H_{1,n}$. The yield prediction errors of $\sim 5.5\%$ reported in trials $H_{1,n}$, $H_{1,(n+af)}$, and $H_{(1+2),n}$, are comparable with other state-of-the-art yield prediction methods, such as those presented by Linker (2018b, 2017), Payne et al. (2014) and Zhou et al. (2012), which reported yield prediction errors of between 10% and 16%. However, while Linker (2018b, 2017) and Payne et al. (2014) used night-time images to prevent detection errors due to natural lighting, the presented methodology is not affected by lighting conditions.

Another advantage of using a MTLs system compared to other devices or systems used for fruit detection is that it allows simultaneous yield monitoring and geometric characterization. The system was able to measure canopy geometrical parameters at the same time, namely height, width, cross-section area and leaf area. Therefore, spatial maps of different canopy features can be created, observing the relationship between canopy structure and fruit tree productivity. This capability makes the designed system a very interesting tool that can be used to analyze the behavior of different fruit tree varieties in relation to its potential of production (Kühn et al., 2003), the fruit location and, especially, how pruning techniques and training systems could affect production depending on the structural organization (Martin-Gorriz et al., 2014).

5. Conclusions

This work presents an analysis of different methodologies based on the use of a mobile terrestrial laser scanner for remote fruit detection and plant geometrical characterization. In order to minimize fruit occlusions, two different approaches were tested: forced air flow and multi-view sensing. The main contributions of this paper were: (1) A methodology for simultaneous fruit location and canopy geometric characterization; (2) An analysis of the usefulness of forced air flow and multi-view approaches for fruit detection, yield prediction and canopy geometric characterization. The results show that the system was able to detect and locate more than 80% of the total annotated fruits and to predict the yield with an RMSE lower than 6%. These fruit detection results are comparable with those obtained with other state-of-the-art methodologies, with the advantages that the presented system is not affected by lighting conditions and also provides geometric characterization of the tree crop, allowing the comparison between yield and canopy structure. From the scanning conditions analysis, it is concluded that the best configuration for yield prediction and geometric characterization

corresponds to mounting the sensor at half the maximum tree height H_l and scanning without forced air flow action. However, if the LiDAR-based system is used for fruit detection, combining data acquired with and without forced air flow action $H_{l,(n+af)}$ or using the multi-view $H_{(1+2),n}$ approach are good options to increase the percentage of fruits detected. If the scanning system is used for both fruit detection and geometric characterization, the best option is the multi-view approach, since it increases the fruit detection rate without excessively penalizing geometric characterization. Future works should focus in the analysis of fruit occlusions in different training systems and extending the present study to other fruit varieties.

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